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Web Technology

Quality Laboratory Manual

ExperimentNo. 07

Program to Implement PHP Variables and Arrays



Course Instructor–

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ASSISTANT PROFESSOR

ExperimentNo.07

Title of Experiment: Program to Implement PHP Variables and Arrays

Aim of Experiment: To understand and implement PHP variables and arrays for storing and manipulating data in a web application.

System Requirements –

Operating System: Windows 10 or above

RAM: Minimum 4 GB

Software/s Needed for Experiment-

- Web Browser
- Text Editor / VS Code
- XAMPP / WAMP Server
- PHP Installed

Experiment Objectives:

- To understand the concept of XSLT.
- To learn how XML data is transformed into HTML.
- To write an XSLT stylesheet.
- To link XML with XSLT.
- To display XML data in structured HTML format.

Experiment Outcomes:

- Understand how PHP variables are declared and used.
- Implement indexed arrays and associative arrays.
- Use loops to display array elements dynamically.
- Develop simple PHP programs for handling data.

Theory:

PHP is a server-side scripting language widely used for web development. It allows developers to create dynamic web pages. PHP Variables

A variable is used to store data values. In PHP, variables start with the \$ symbol followed by the variable name.

Example:

```
$studentName = "Rahul";  
$rollNo = 101;
```

PHP supports different data types such as:

- Integer

- Float
- String
- Boolean
- Array
- Object

PHP Arrays

An array is used to store multiple values in a single variable.

Types of Arrays in PHP

1. Indexed Array – Uses numeric index.
2. Associative Array – Uses named keys.

Example:

```
$marks = array(85, 90, 78);
```

Example of Associative Array:

```
$student = array("Name"=>"Rahul", "RollNo"=>101, "Branch"=>"CSE");
```

Arrays are commonly used to store lists of values such as student data, product details, etc.

Program

Program 1: PHP Variables

```
<?php
$name = "Amruta";
$rollno = 101;
$branch = "Computer Engineering";

echo "<h2>Student Information</h2>";
echo "Name: " . $name . "<br>";
echo "Roll No: " . $rollno . "<br>";
echo "Branch: " . $branch;
?>
```

Program 2: PHP Indexed Array

```
<?php
$marks = array(85, 90, 78, 88);

echo "<h2>Student Marks</h2>";

for($i=0; $i<count($marks); $i++)
{
    echo "Marks: " . $marks[$i] . "<br>";
}
?>
```

Program 3: PHP Associative Array

```
<?php
$student = array(
    "Name"=>"Rahul",
    "RollNo"=>102,
    "Branch"=>"Information Technology"
);

echo "<h2>Student Details</h2>";

foreach($student as $key => $value)
{
    echo $key . " : " . $value . "<br>";
}
?>
```

Task for Students: create a PHP program for a Student Information System using variables and arrays.

Task Requirements:

- Declare PHP variables to store student information (Name, Roll No, Branch).
- Create an indexed array to store marks of at least 5 subjects.
- Create an associative array to store student details.
- Use loops (for / foreach) to display array values.
- Display the output in a structured HTML format.

Submission Requirement:

- PHP Program File
- Screenshot of Program Code
- Screenshot of Output in Browser

Observations:

In this experiment, PHP variables were used to store student information. Indexed arrays were used to store subject marks, and associative arrays were used to store key-value pairs such as student details. Loops like **for** and **foreach** were used to display array values dynamically on the web page. The program executed successfully and displayed the student information in the browser.

Conclusion:

This experiment demonstrated the implementation of PHP variables and arrays. Students learned how to store multiple values using arrays and display them using loops in PHP, which is essential for developing dynamic web applications.

Expected Oral Questions:

1. What is PHP?
2. What is a PHP variable?
3. What are the rules for naming variables in PHP?
4. What is an array in PHP?
5. What are the types of arrays in PHP?
6. What is the difference between indexed array and associative array?
7. What is the use of the foreach loop?
8. What is the difference between echo and print in PHP?

FAQs in Interview:

1. What is the use of arrays in PHP?
2. What is the difference between indexed and associative arrays?
3. What is the syntax for declaring an array in PHP?
4. What is count() function in PHP?
5. Can arrays store different data types in PHP?
6. What is the difference between for loop and foreach loop?
7. What is a multidimensional array?
8. How are arrays used in real-world web applications?